

LOGIC No: SEQ-6701	DESCRIPTION: LP SEPARATOR LEVEL PROTECTION (FA-6710 / LV-6701) LOGIC	SIL: SIL 1	TAG: LV-6701 FLOC: TJC-LLD-6700-01
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CAUSE & EFFECT MATRIX

ID	INITIATOR / CAUSE (INPUT)	TAG	SET POINT	VOTE	EFF-1 OPEN LV-6701 FULLY (relieve level)	EFF-2 CLOSE LV-6701 (prevent gas blow-by)	EFF-3 VENT ACTUATOR - LV-6701 FAILS CLOSED	EFF-4 ANNUNCIATE ALARM ON DCS
T1	Separator level HIGH-HIGH	LSHH-6710	> 85 %	1oo2	X			X
T2	Separator level LOW-LOW	LSLL-6710	< 15 %	1oo2		X		X
T3	ESD activation (manual)	HS-6701	ESD signal	1oo1			X	X
T4	Instrument air pressure LOW	PSL-6702	< 1.0 barg	1oo1			X	X

Legend: X = initiator (row) actuates that effect (column). Voting: 1oo1/1oo2/2oo3 = number of transmitters that must agree to trip.

EFFECT ID	FINAL ELEMENT / ACTION
EFF-1	OPEN LV-6701 FULLY (relieve level)
EFF-2	CLOSE LV-6701 (prevent gas blow-by)
EFF-3	VENT ACTUATOR - LV-6701 FAILS CLOSED
EFF-4	ANNUNCIATE ALARM ON DCS

#	START PERMISSIVE (AND-gate)	SIGNAL
1	Instrument air available (> 1.2 barg)	PI-6702
2	Positioner DVC6200 in AUTO/healthy	DVC6200
3	Bypass HV-6701 closed (normal operation)	HV-6701
=>	ALL permissives TRUE (AND) => START LV-6701 ENABLED	RUN

NOTES: 1. Trip set points are training values. 2. On any trip the effects marked X are actuated by the Safety PLC and annunciated on the DCS. 3. A trip is latched and requires a manual reset once the cause has cleared and permissives are healthy. 4. Refer to P&ID TJC-LLD-PID-6701 for instrument loops and the Datasheet for design limits of LV-6701.